

Bhagyashree

Role: MIT App & Website Developer

Project Explanation:

I, Bhagyashree, developed the MIT App and project website for the Smart Multi-Control Robotic Car. I designed the mobile application for Bluetooth-based control and created the project website using HTML, CSS, and JavaScript to present project information and features. I also tested the communication between the application and the robotic car for smooth operation.

Responsibilities:

- Developed the MIT App Inventor application
- Designed Bluetooth connection interface
- Created Forward, Backward, Left, Right, and Stop buttons
- Developed Home, School, Office, and Hospital route buttons
- Added speed control options in the application
- Created the project website using HTML, CSS, and JavaScript
- Designed website pages and user interface
- Tested app and website functionality
- Assisted in project demonstration and presentation activities

Shivangouda

Role: Hardware Integration Engineer

Project Explanation:

I, Shivangouda, was responsible for assembling the robotic car and integrating all hardware components. I ensured proper wiring, power distribution, and smooth functioning of the motors and electronic modules.

Responsibilities:

- Assembled the 4WD robotic car chassis
- Mounted BO motors and wheels
- Connected ESP32, motor driver, and battery
- Managed complete wiring and circuit connections
- Installed and tested power supply system
- Verified motor movement in all directions
- Assisted in overall hardware testing and maintenance

Chetan

Role: Sensor & Automation Developer

Project Explanation:

I, Chetan, worked on the automation features of the robotic car. I integrated the ultrasonic sensor and servo motor to enable obstacle detection and autonomous navigation.

Responsibilities:

- Connected ultrasonic sensor and servo motor
- Developed obstacle detection mechanism
- Implemented automatic obstacle avoidance system
- Tested distance measurement accuracy
- Assisted in autonomous navigation development
- Verified sensor performance under different conditions
- Improved safety and navigation efficiency of the robot

Supriya

Role: Documentation & Presentation Coordinator

Project Explanation:

I, Supriya, handled the project documentation and presentation activities. I prepared reports, recorded project results, and presented the objectives, applications, and future scope of the robotic car project.

Responsibilities:

- Prepared project report and documentation
- Created presentation slides and posters
- Recorded testing observations and results
- Documented project applications and advantages
- Explained future scope and enhancements
- Coordinated final project presentation
- Assisted in project demonstration and evaluation

SMART MULTI-CONTROL ROBOTIC CAR USING ESP32

AIM

To develop a Bluetooth-controlled Smart Robotic Car using ESP32 technology capable of manual control, autonomous obstacle avoidance, speed regulation, and route-based navigation.

INTRODUCTION

The Smart Multi-Control Robotic Car is an advanced robotic vehicle developed using ESP32, L293N motor driver, ultrasonic sensor, servo motor, and a mobile application. The project demonstrates the concepts of robotics, automation, wireless communication, and embedded systems. The robotic car can be controlled manually through a Bluetooth-enabled mobile application and can also operate automatically by detecting and avoiding obstacles. The project is designed to provide an efficient, user-friendly, and intelligent robotic system for educational and automation applications.

OBJECTIVES

- To design and develop a Bluetooth-controlled robotic car using ESP32.
- To implement manual movement control through a mobile application.
- To develop an autonomous obstacle detection and avoidance system.
- To provide speed control functionality for smooth operation.
- To implement predefined route navigation such as Home, School, Office, and Hospital.
- To demonstrate the application of embedded systems and robotics.

APPARATUS AND SYSTEM REQUIREMENTS

Hardware:

ESP32 Development Board, L293N Motor Driver, HC-SR04 Ultrasonic Sensor, SG90 Servo Motor, 4 BO Motors, 4WD Chassis, Battery Pack, Wheels, Zero PCB, and Connecting Wires.

Software:

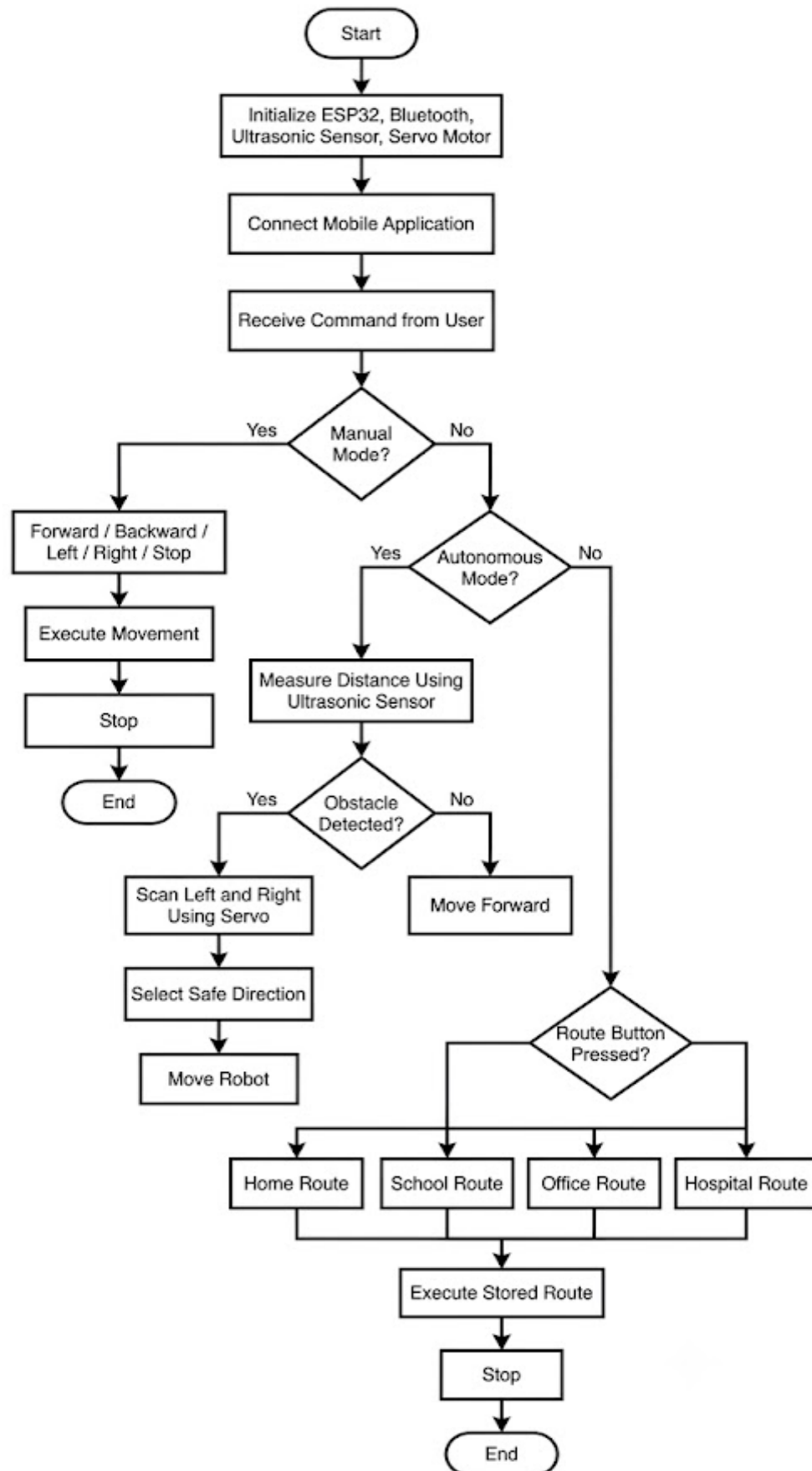
Arduino IDE, ESP32 Board Package, MIT App Inventor, Embedded C Programming, HTML, CSS, and JavaScript.

WORKING PRINCIPLE

The robotic car is controlled through a mobile application connected to the ESP32 using Bluetooth communication. When the user presses control buttons, commands are transmitted to the ESP32, which controls the motor driver and motors to perform various movements. In autonomous mode, the ultrasonic sensor continuously measures the distance of nearby objects. When an obstacle is detected, the servo motor rotates the sensor to scan different directions and identify a safe path. The system also includes speed control and predefined route navigation functions such as Home, School, Office, and Hospital.

FLOWCHART

'Smart Multi-Control Robotic Car Using ESP32'



ADVANTAGES

- Wireless Bluetooth control
- Autonomous obstacle avoidance
- Easy and user-friendly operation
- Adjustable speed control
- Low-cost implementation
- Suitable for educational and research applications
- Supports predefined route navigation

DISADVANTAGES

- Limited Bluetooth communication range
- Accuracy depends on sensor performance
- Battery backup is limited
- Cannot perform advanced navigation without GPS
- Performance may vary on uneven surfaces

CONCLUSION

The project successfully demonstrates manual control, autonomous navigation, and obstacle avoidance using ESP32 technology. The developed system is reliable, efficient, and capable of intelligent movement in different environments.

FUTURE SCOPE

- Wi-Fi-based remote control
- GPS navigation system
- Live camera monitoring
- Voice-controlled operation
- IoT-based monitoring and control
- AI-based autonomous navigation
- Object tracking and follow-me mode
- Smart home and industrial automation applications